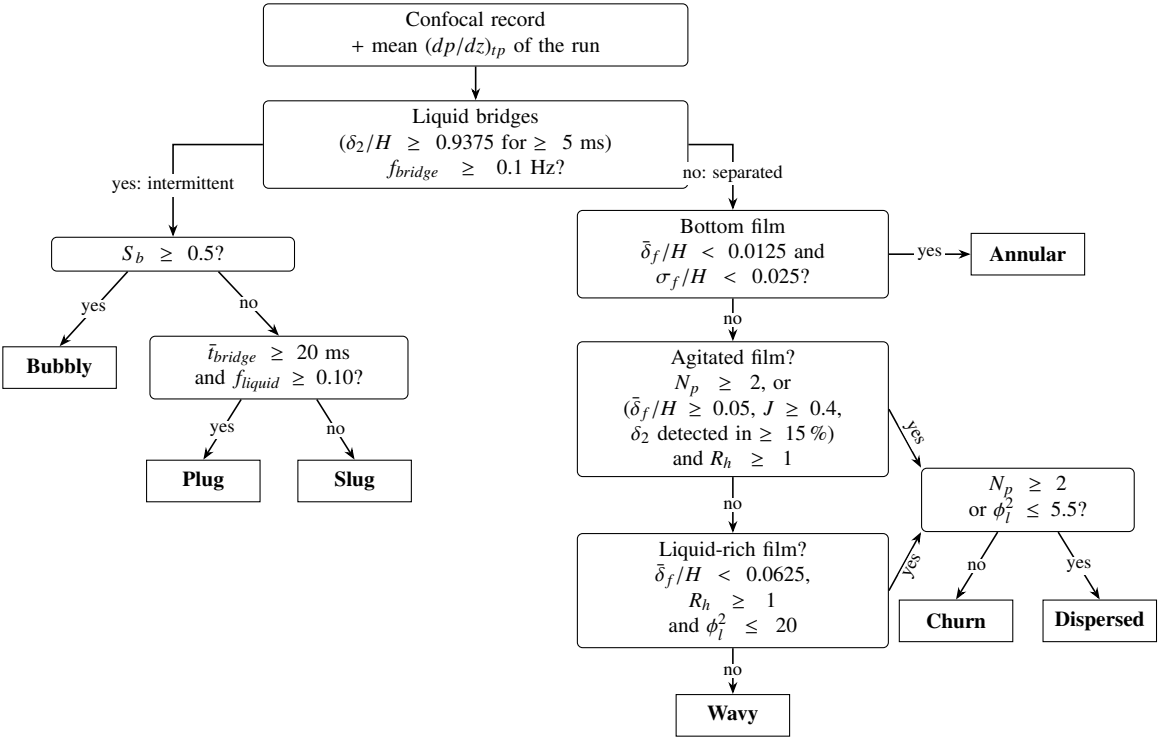
Confocal record
+ mean $(dp/dz)_{tp}$ of the run
Liquid bridges
($\delta_2/H \geq 0.9375$ for ≥ 5 ms)
$f_{bridge} \geq 0.1$ Hz?
yes: intermittent
no: separated
$S_b \geq 0.5$?
yes
no
Bubbly
$\bar{t}_{bridge} \geq 20$ ms
and $f_{liquid} \geq 0.10$?
yes
no
Plug
Slug
Bottom film
$\bar{\delta}_f/H < 0.0125$ and
$\sigma_f/H < 0.025$?
yes
Annular
no
Agitated film?
$N_p \geq 2$, or
($\bar{\delta}_f/H \geq 0.05$, $J \geq 0.4$,
δ_2 detected in $\geq 15\,\%$)
and $R_h \geq 1$
yes
$N_p \geq 2$
or $\phi_l^2 \leq 5.5$?
no
yes
Churn
Dispersed
no
Liquid-rich film?
$\bar{\delta}_f/H < 0.0625$,
$R_h \geq 1$
and $\phi_l^2 \leq 20$
yes
no
Wavy